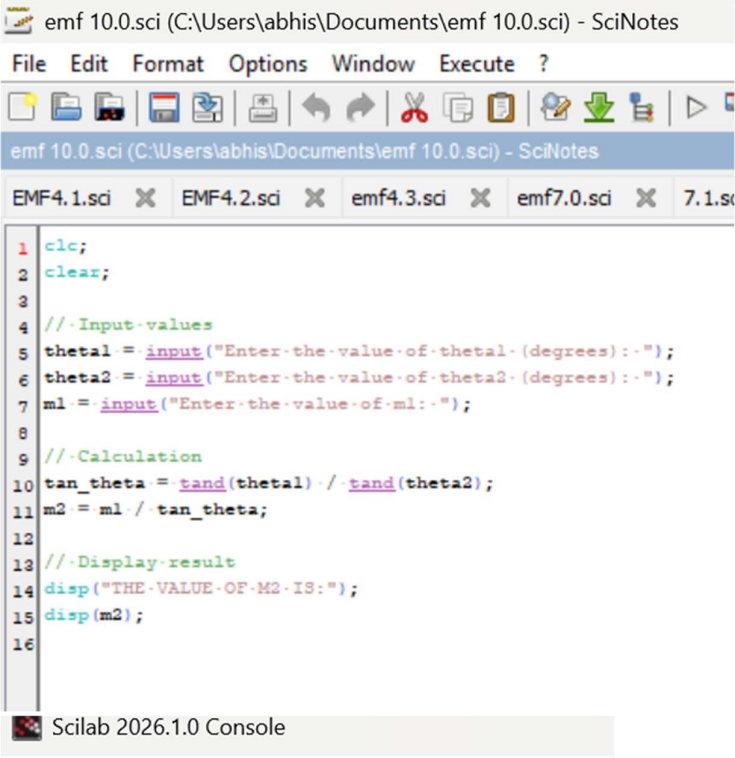


Exp no:10 Implement Boundary condition between two different dielectric medium in Magnetic Field and find the relative permeability of the different medium.

Aim: To write program for Boundary condition between two different dielectric media in magnetic field using SCILAB.

Software required:

- SCILAB version 6.0.1



The screenshot shows the Scilab 2026.1.0 IDE. The main window displays a script named 'emf 10.0.sci'. The script contains the following code:

```
1 clc;
2 clear;
3
4 //Input-values
5 theta1 = input("Enter the value of theta1 (degrees) :- ");
6 theta2 = input("Enter the value of theta2 (degrees) :- ");
7 m1 = input("Enter the value of m1 :- ");
8
9 //Calculation
10 tan_theta = tand(theta1) / tand(theta2);
11 m2 = m1 / tan_theta;
12
13 //Display result
14 disp("THE VALUE OF M2 IS:");
15 disp(m2);
16
```

Below the script editor, the Scilab 2026.1.0 Console is visible, showing the execution output:

Enter the value of theta1 (degrees): 50

Enter the value of theta2 (degrees): 75

Enter the value of m1: 5

"THE VALUE OF M2 IS:"  
15.657812

-->

(10) Given

$$\theta_1 = 50^\circ$$

$$\theta_2 = 75^\circ$$

$$m_1 = 5$$

Formula  $\tan(\theta) = \frac{\sin(\theta)}{\cos(\theta)}$

$$\tan 50^\circ = 1.1918$$

$$\tan 75^\circ = 3.7321$$

$$\tan 35^\circ = \frac{\text{opposite}}{\text{adjacent}} = \frac{\sin(35^\circ)}{\cos(35^\circ)} = 0.7002$$

o/p

$$\frac{\tan 50^\circ}{\tan 75^\circ} = \frac{1.1918}{3.7321} = 0.3193$$

$$m_2 = m_1 \times 0.3193 = 5 \times 0.3193 = 1.596578$$

$m_2 = 1.596578$

Case 1: Consider two identical media in the presence of a magnetic field. The relative permeability in both media is 1. What will be the effect of this on the magnetic field and flux density across the boundary?

Analytical Calculation

Given:

1. Two identical media with  $\mu_r = 1$ .
2. Magnetic permeability of both media is  $\mu = \mu_0$  is the permeability of free space.
3. Magnetic field intensity in the first medium:  $H_1 = 10 \text{ A/m}$ .
4. The problem asks for the behaviour of the magnetic field  $H$  and flux density  $B$  across the boundary.

Assumptions:

□ The media are in static conditions, with no additional sources or boundary effects (e.g., surface currents or magnetic charges).

Boundary Conditions:

1. Tangential component of  $\mathbf{H}$ :

$$H_{1t} = H_{2t}$$

The tangential components of  $\mathbf{H}$  are continuous across the boundary.

2. Normal component of  $\mathbf{B}$ :

$$B_{1n} = B_{2n}$$

The normal components of  $\mathbf{B}$  are continuous across the boundary.

3. Relationship between  $\mathbf{B}$  and  $\mathbf{H}$ :

$$\mathbf{B} = \mu \mathbf{H}$$

Since  $\mu = \mu_0$ ,  $\mathbf{B} = \mu_0 \mathbf{H}$  in both media.

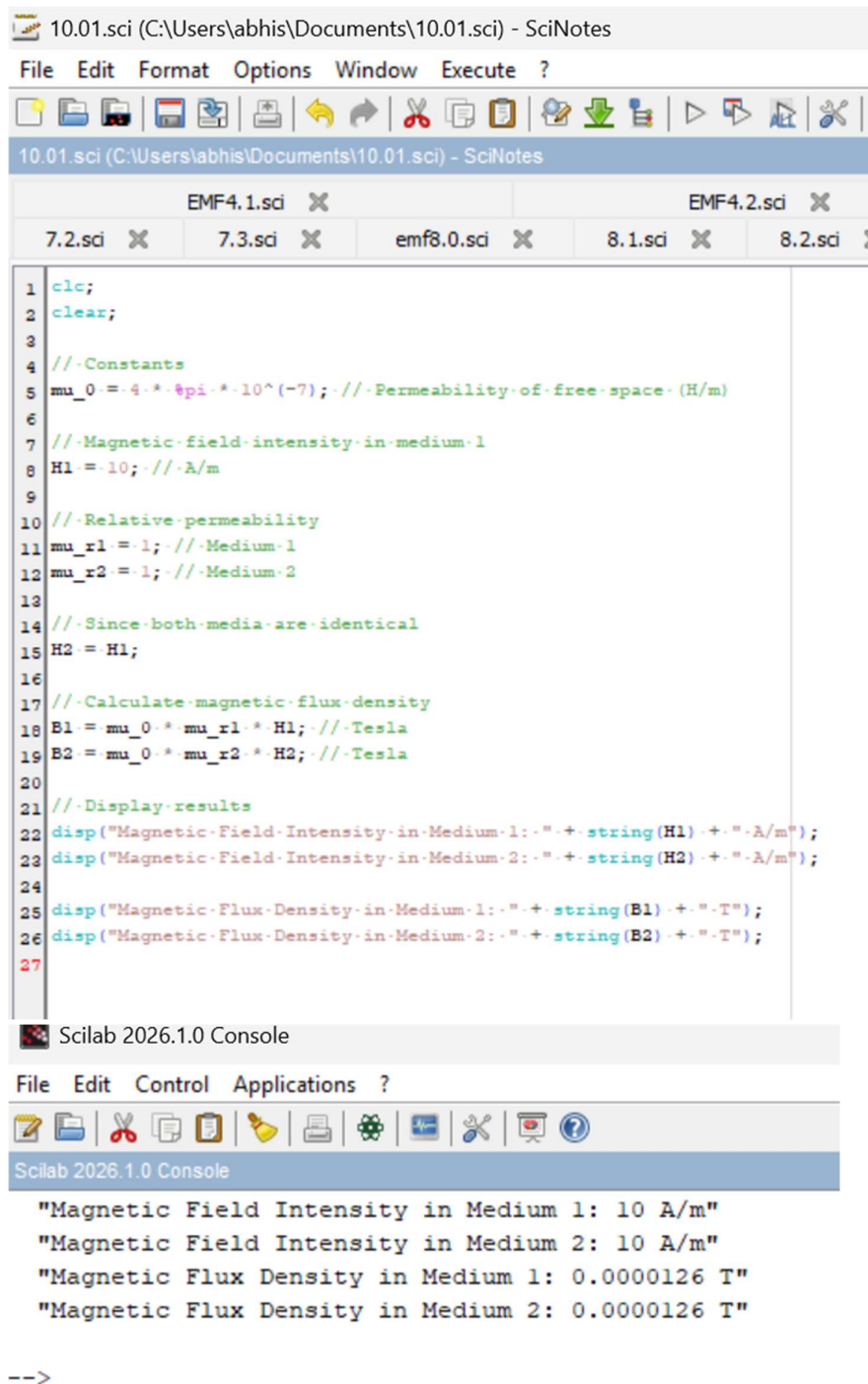
Calculation of  $B_1$  and  $B_2$ :

□ In the first medium:

$$B_1 = \mu_0 H_1 = (4\pi \times 10^{-7}) \times 10 = 4\pi \times 10^{-6} \text{ T}$$

□ Since the media are identical, and boundary conditions ensure continuity:

$$H_2 = H_1 = 10 \text{ A/m}, B_2 = B_1 = 4\pi \times 10^{-6} \text{ T}$$



The image shows a screenshot of a computer screen with two windows. The top window is titled "10.01.sci (C:\Users\abhis\Documents\10.01.sci) - SciNotes" and contains a MATLAB script. The script defines constants for permeability and magnetic field intensity, calculates magnetic flux density for two media, and displays the results. The bottom window is titled "Scilab 2026.1.0 Console" and shows the output of the script, which is the same as the display commands in the script.

```
1 clc;
2 clear;
3
4 //Constants
5 mu_0 = 4 * pi * 10^(-7); // Permeability of free space (H/m)
6
7 //Magnetic field intensity in medium 1
8 H1 = 10; // A/m
9
10 //Relative permeability
11 mu_r1 = 1; // Medium 1
12 mu_r2 = 1; // Medium 2
13
14 //Since both media are identical
15 H2 = H1;
16
17 //Calculate magnetic flux density
18 B1 = mu_0 * mu_r1 * H1; // Tesla
19 B2 = mu_0 * mu_r2 * H2; // Tesla
20
21 //Display results
22 disp("Magnetic Field Intensity in Medium 1: " + string(H1) + " A/m");
23 disp("Magnetic Field Intensity in Medium 2: " + string(H2) + " A/m");
24
25 disp("Magnetic Flux Density in Medium 1: " + string(B1) + " T");
26 disp("Magnetic Flux Density in Medium 2: " + string(B2) + " T");
27
```

```
"Magnetic Field Intensity in Medium 1: 10 A/m"
"Magnetic Field Intensity in Medium 2: 10 A/m"
"Magnetic Flux Density in Medium 1: 0.0000126 T"
"Magnetic Flux Density in Medium 2: 0.0000126 T"
```

Result :

Thus the effect of two identical media on the magnetic field and flux density across the boundary is calculated.

Case 2: In a system with two media, where the relative permeability of medium 1 is  $\mu_{r1}=4$  and medium

2 is a vacuum ( $\mu_{r2}=1$ ), how would the magnetic field intensity and flux density behave at the boundary?

Analytical Calculation:

**Given:**

1. Medium 1 has relative permeability  $\mu_{r1} = 4$ .
2. Medium 2 is a vacuum with relative permeability  $\mu_{r2} = 1$ .
3. Magnetic field intensity in medium 1:  $H_1 = 10 \text{ A/m}$ .
4. We need to determine how **H** and **B** behave at the boundary between the two media.

**Properties:**

1. Permeability of medium 1:

$$\mu_1 = \mu_0 \mu_{r1} = 4\mu_0$$

2. Permeability of medium 2 (vacuum):

$$\mu_2 = \mu_0$$

3. Magnetic flux density (**B**) is related to **H** by:

$$\mathbf{B} = \mu \mathbf{H}$$

**Boundary Conditions:**

1. Tangential component of **H**:

$$H_{1t} = H_{2t}$$

The tangential components of **H** are continuous across the boundary.

2. Normal component of **B**:

$$B_{1n} = B_{2n}$$

The normal components of **B** are continuous across the boundary.

**Step 1: Magnetic Flux Density ( $B_1$  in Medium 1)**

In medium 1:

$$B_1 = \mu_1 H_1 = (4\mu_0) \cdot 10 = 40\mu_0 \text{ T.}$$

**Step 2: Tangential Component of  $H$  in Medium 2 ( $H_2$ )**

Since the tangential components of **H** are continuous across the boundary:

#### Step 4: Normal Component of $B$

The normal components of  $\mathbf{B}$  are continuous:

$$B_{1n} = B_{2n}.$$

From the values calculated:

- $B_1 = 40\mu_0 \text{ T}$
- $B_2 = 10\mu_0 \text{ T}$

10.2.sci (C:\Users\abhis\Documents\10.2.sci) - SciNotes

File Edit Format Options Window Execute ?

10.2.sci (C:\Users\abhis\Documents\10.2.sci) - SciNotes

EMF4.1.sci X

EMF4.2.sci X

7.3.sci X

emf8.0.sci X

8.1.sci X

8.2.sci X

8,3.sci X

```

1  clc;
2  clear;
3
4  //Constants
5  mu_0 = 4.*pi.*10^(-7); // Permeability of free space (H/m)
6
7  //Magnetic field intensity in medium 1
8  H1 = 10; // A/m
9
10 //Relative permeability of the media
11 mu_r1 = 4; // Medium 1
12 mu_r2 = 1; // Medium 2 (vacuum)
13
14 //Boundary condition
15 //mu1.*H1 = mu2.*H2
16 H2 = (mu_r1.*H1) ./ mu_r2;
17
18 //Calculate magnetic flux density
19 B1 = mu_0.*mu_r1.*H1;
20 B2 = mu_0.*mu_r2.*H2;
21
22 //Display results
23 disp("Magnetic Field Intensity in Medium 1: " + string(H1) + " A/m");
24 disp("Magnetic Field Intensity in Medium 2: " + string(H2) + " A/m");
25
26 disp("Magnetic Flux Density in Medium 1: " + string(B1) + " T");
27 disp("Magnetic Flux Density in Medium 2: " + string(B2) + " T");
28

```

Scilab 2026.1.0 Console

File Edit Control Applications ?

Scilab 2026.1.0 Console

```

"Magnetic Field Intensity in Medium 1: 10 A/m"
"Magnetic Field Intensity in Medium 2: 40 A/m"
"Magnetic Flux Density in Medium 1: 0.0000503 T"
"Magnetic Flux Density in Medium 2: 0.0000503 T"

```

-->

Result :

Thus the effect of two different media on the magnetic field and flux density across the boundary is calculated.

Case 3: If medium 1 has a relative permeability  $\mu_{r1}=0.3$  and medium 2 has a relative permeability  $\mu_{r2}=2$ , calculate the magnetic field intensity in both media at the boundary if the magnetic field intensity in medium 1 is known to be  $H_1=10$  A/m. Analytical Calculations:

---

**Given:**

**1. Relative permeability:**

- Medium 1:  $\mu_{r1} = 0.3$
- Medium 2:  $\mu_{r2} = 2$

**2. Magnetic field intensity in medium 1:  $H_1 = 10$  A/m.**

**3. The permeability in each medium is:**

$$\mu_1 = \mu_0 \mu_{r1}, \quad \mu_2 = \mu_0 \mu_{r2}.$$

**4. Relationship between  $\mathbf{B}$  and  $\mathbf{H}$ :**

$$\mathbf{B} = \mu \mathbf{H}.$$

**Step 1: Calculate  $B_1$  in Medium 1**

$$B_1 = \mu_1 H_1 = (\mu_0 \mu_{r1}) H_1 = (\mu_0 \cdot 0.3) \cdot 10 = 3\mu_0 \text{ T}.$$

**Step 2: Tangential Component of  $H$  in Medium 2**

Since the tangential component of  $H$  is continuous:

$$H_2 = H_1 = 10 \text{ A/m}.$$

**Step 3: Magnetic Flux Density  $B_2$  in Medium 2**

$$B_2 = \mu_2 H_2 = (\mu_0 \mu_{r2}) H_2 = (\mu_0 \cdot 2) \cdot 10 = 20\mu_0 \text{ T}.$$

**Step 4: Verify Normal Component of  $B$**

The normal components of  $B$  are continuous:

$$B_{1n} = B_{2n}.$$



10.03.sci (C:\Users\abhis\Documents\10.03.sci) - SciNotes

File Edit Format Options Window Execute ?

10.03.sci (C:\Users\abhis\Documents\10.03.sci) - SciNotes

EMF4.1.sci X EMF4.2.sci X

emf8.0.sci X 8.1.sci X 8.2.sci X 8,3.sci X emf9.0.sci

```

1  clc;
2  clear;
3
4  //Constants
5  mu_0 = 4.*pi.*10^(-7); // Permeability of free space (H/m)
6
7  //Magnetic field intensity in medium 1
8  H1 = 10; // A/m
9
10 //Relative permeability for the media
11 mu_r1 = 0.3; // Medium 1
12 mu_r2 = 2; // Medium 2
13
14 //Calculate magnetic field intensity in medium 2
15 H2 = (mu_r1./mu_r2).*H1;
16
17 //Calculate magnetic flux density
18 B1 = mu_0.*mu_r1.*H1;
19 B2 = mu_0.*mu_r2.*H2;
20
21 //Display results
22 disp("Magnetic Field Intensity in Medium 1: " + string(H1) + " A/m");
23 disp("Magnetic Field Intensity in Medium 2: " + string(H2) + " A/m");
24
25 disp("Magnetic Flux Density in Medium 1: " + string(B1) + " T");
26 disp("Magnetic Flux Density in Medium 2: " + string(B2) + " T");
27

```

Scilab 2026.1.0 Console

File Edit Control Applications ?

Scilab 2026.1.0 Console

```

"Magnetic Field Intensity in Medium 1: 10 A/m"
"Magnetic Field Intensity in Medium 2: 1.5 A/m"
"Magnetic Flux Density in Medium 1: 0.0000038 T"
"Magnetic Flux Density in Medium 2: 0.0000038 T"

```

-->

Result :Thus the effect of two different media on the magnetic field and flux density across the boundary is calculated.